

# Метод Пуанкаре

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In[211]:=

```
ClearAll["Global`*"]
```

## Условие задачи

$$x_{tt} + \omega^2 x = \epsilon (1 - x^2) x_t - \epsilon x^3$$

In[212]:=

```
Eq = x''[t] + \omega^2 * x[t] == \epsilon * (1 - x[t]^2) * x'[t] - \epsilon * x[t]^3
```

Out[212]=

$$\omega^2 x[t] + x''[t] == -\epsilon x[t]^3 + \epsilon (1 - x[t]^2) x'[t]$$

## Разложение в ряд

In[213]:=

```
\Omega = 1 + \Omega1 * \epsilon;  
\xi = \xi0[\tau] + \xi1[\tau] * \epsilon;
```

In[215]:=

```
T2Tau = {t \to \tau * \Omega / \omega, x[t] \to \xi, x'[t] \to D[\xi, \tau] * \omega / \Omega, x''[t] \to D[\xi, {\tau, 2}] * (\omega / \Omega)^2}
```

Out[215]=

$$\left\{ t \rightarrow \frac{\tau (1 + \epsilon \Omega 1)}{\omega}, x[t] \rightarrow \xi 0[\tau] + \epsilon \xi 1[\tau], x'[t] \rightarrow \frac{\omega (\xi 0'[\tau] + \epsilon \xi 1'[\tau])}{1 + \epsilon \Omega 1}, x''[t] \rightarrow \frac{\omega^2 (\xi 0''[\tau] + \epsilon \xi 1''[\tau])}{(1 + \epsilon \Omega 1)^2} \right\}$$

In[216]:=

```
Eq1 = Simplify[MultiplySides[Eq /. T2Tau, Ω^2, Assumptions→{Ω > 0}]]
```

Out[216]=

$$\omega^2 \left( (1 + \epsilon \Omega^1)^2 (\xi_0[\tau] + \epsilon \xi_1[\tau]) + \xi_0''[\tau] + \epsilon \xi_1''[\tau] \right) = \\ \epsilon (1 + \epsilon \Omega^1) \left( - \left( (1 + \epsilon \Omega^1) (\xi_0[\tau] + \epsilon \xi_1[\tau])^3 \right) + \omega \left( 1 - (\xi_0[\tau] + \epsilon \xi_1[\tau])^2 \right) (\xi_0'[\tau] + \epsilon \xi_1'[\tau]) \right)$$

## Нулевое приближение

In[217]:=

```
Eqξ0 = Simplify[CoefficientList[Eq1[[1]], ε] [[1]] == CoefficientList[Eq1[[2]], ε] [[1]], Assumptions→{ω > 0}]
```

Out[217]=

$$\xi_0[\tau] + \xi_0''[\tau] == 0$$

In[218]:=

```
Solξ0 = DSolve[{Eqξ0, ξ0'[0] == 0, ξ0[0] == M}, ξ0[τ], τ] [[1]]
```

Out[218]=

$$\{\xi_0[\tau] \rightarrow M \cos[\tau]\}$$

## Первое приближение

In[219]:=

```
Eqξ1 = Simplify[Expand[(CoefficientList[Eq1[[1]], ε] [[2]] == CoefficientList[Eq1[[2]], ε] [[2]]) /. {Solξ0[[1]], D[Solξ0[[1]], τ]}]]
```

Out[219]=

$$M^3 \cos[\tau]^3 + M \omega \sin[\tau] + \omega^2 (2 M \Omega^1 \cos[\tau] + \xi_1[\tau] + \xi_1''[\tau]) = M^3 \omega \cos[\tau]^2 \sin[\tau]$$

In[220]:=

```
Solξ1 = DSolve[{Eqξ1, ξ1'[0] == 0, ξ1[0] == 0}, ξ1[τ], τ][[1]]
```

Out[220]=

$$\left\{ \xi_1[\tau] \rightarrow \frac{1}{32 \omega^2} \left( 9 M^3 \cos[\tau] + 16 M \tau \omega \cos[\tau] - 4 M^3 \tau \omega \cos[\tau] + 32 M \omega^2 \Omega_1 \cos[\tau] - 8 M^3 \cos[\tau]^3 - 32 M \omega^2 \Omega_1 \cos[\tau]^3 - M^3 \cos[\tau] \times \cos[4 \tau] - \right. \right. \\ \left. \left. 12 M^3 \tau \sin[\tau] - 16 M \omega \sin[\tau] + 9 M^3 \omega \sin[\tau] - 32 M \tau \omega^2 \Omega_1 \sin[\tau] + 16 M \omega \cos[\tau]^2 \sin[\tau] - 8 M^3 \omega \cos[\tau]^2 \sin[\tau] - M^3 \omega \cos[4 \tau] \times \sin[\tau] - \right. \right. \\ \left. \left. 8 M \omega \cos[\tau] \times \sin[2 \tau] - 8 M^3 \sin[\tau] \times \sin[2 \tau] - 16 M \omega^2 \Omega_1 \sin[\tau] \times \sin[2 \tau] + M^3 \omega \cos[\tau] \times \sin[4 \tau] - M^3 \sin[\tau] \times \sin[4 \tau] \right) \right\}$$

## Искомая поправка

In[221]:=

```
Sol = Solve[Coefficient[ξ1[τ] /. Solξ1, τ * Cos[τ]] == 0 && Coefficient[ξ1[τ] /. Solξ1, τ * Sin[τ]] == 0, {M, Ω1}, Assumptions→{M > 0, ω > 0}]
```

Out[221]=

$$\left\{ M \rightarrow 2, \Omega_1 \rightarrow -\frac{3}{2 \omega^2} \right\}$$

In[222]:=

```
x1 = ((ξ0[τ] /. Solξ0) /. {τ → t * ω / (1 + ε * Ω1)}) /. Sol;
Print["x(t) = ", TraditionalForm[x1]]
```

$$x(t) = 2 \cos \left( \frac{t \omega}{1 - \frac{3 \epsilon}{2 \omega^2}} \right)$$

## Сравнение с точным решением

In[224]:=

```

nx1 = x1 /. { $\omega \rightarrow 1$ ,  $\epsilon \rightarrow 0.1$ };
nx0 = 2 * Cos[t];
nSol = NDSolve[{Eq /. { $\omega \rightarrow 1$ ,  $\epsilon \rightarrow 0.1$ }, x'[0] == 0, x[0] == 2}, x, {t, 0, 10}];
nx = Evaluate[x[t] /. nSol];

```

In[228]:=

```

Plot[{nx0, nx1, nx}, {t, 0, 10}, PlotLegends -> Placed[{"Без поправки", "С поправкой", "Точное решение"}, Right], AxesLabel -> {t, x[t]},
GridLines -> Automatic, PlotStyle -> {Blue, {Blue, DotDashed}, Red}, PlotRange -> All]

```

Out[228]=

